Computing with Space

Distributed Systems

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1. Prologue
2. Space in Math & Logic
3. Space in Computer Science

Next in Line...

Space in Distributed Computing

- *distribution* is first of all physical distribution, which basically means *spatial distribution*
- as in the case of time, we then need to understand what is space, what we mean when we think and speak of space
- how we do *represent* space in our *reasoning* and in *computing* as well
- nowadays, a huge amount of application scenarios require *spatial reasoning* and *spatial computing*

Disclaimer

- the following slides borrow a good deal from the tutorial "Spatial Multi-Agent Systems" held at the 18th European Agent Systems Summer School (EASSS 2016)
- original slides can be found at
  http://www.slideshare.net/andreaomicini/spatial-multiagent-systems
- developed with the help of Stefano Mariani and Mirko Viroli

Next in Line...

Humans & Space

Environment awareness, measure & modelling

For early humans

- awareness of the surrounding environment as the premise to self-awareness[Martelet, 1998]
- modelling and measure of the environment as the premises to goal-driven actions

Space & activity

For humans, space is a conceptual tool

- to model the environment where we live
- to organise resources and activities
- and their dynamics as well

Spatial Reasoning I

Representing space

- the way in which we represent space mutually affects the way in which we can *reason about* space
- in some way not yet well understood, probably
  - for instance, it easy to find two clearly contrasting views on the matter explicitly expressed [Li and Gleitman, 2002, Levinson et al., 2002]

Spatial Reasoning II

Reasoning on space

- humans have their own ways to represent space, and to process spatial information[Byrne and Johnson-Laird, 1989]
- spatial reasoning is a feature of the intelligent process—humans are not alone[Gentner, 2007, Haun et al., 2006]
- human spatial reasoning is may be either qualitative or quantitative—but is mostly *approximated*[Dutta, 1988]

! for a well-organised account of how humans (learn to) represent and reason about space see[Clements and Battista, 1992]

Focus on. . .

Geometry

Modelling space

Abstracting away from our perception of reality

- basic geometric *concepts* (point, line, angle, circle, . . .)
- basic geometric *shapes* (triangle, rectangle, trapezoid, . . .)
- from Babylonians to Egyptians to Greeks
- Babylonian *astronomy*
- Greek *geometry*

Euclidean Geometry

Axiomatic approach to geometry

- Euclide's *Elements* [Kline, 1972]
- geometry [Aiello et al., 2012]
  - no longer seen *"as a set of empirical observations and practical methods for measuring distances, area of land, etc."*
  - instead conceived as *"an abstract mathematical theory, which, while rooted in the perceived reality, had nevertheless its own, absolute right of existence and development"*
- representing space and reasoning on space through axioms, theorems, proofs, . . .

Geometry & Space beyond Human Perception

Modelling "imaginary" space

- beyond Euclidean space
  - physical space may not satisfy Euclid's fifth postulate (axiom)
  - *non-euclidean* geometry
  - Riemann (elliptic), Bolyai & Lobachevsky (hyperbolic), ...
- representing *true* physical space beyond our direct sensorial-cognitive perception

[Aiello et al., 2012]

> *That discovery was an unsurpassed manifestation of the superiority of the abstract, logical approach of mathematics over the empirical approach underpinning the natural sciences, at least when it comes to comprehending such fundamental physical concepts as space and time.*

Focus on. . .

From Euclid to Tarski [Aiello et al., 2012] I

Basic question

Morris Kline, Foreword [Russell, 1956]

> *What geometrical knowledge must be the logical starting point for a science of space and must also be logically necessary to the experience of any form of externality?*

Axiomatic investigations of the foundations of geometry

- axiomatic approach to geometry from Euclid to Peano
- studying relationships between axiomatic systems & basic geometric notions
- (sound) *axiomatic* foundation for geometry by Hilbert [Hilbert, 1950]

From Euclid to Tarski[Aiello et al., 2012] II

Analytic method in geometry of space

- Descartes' *coordinatisation* of the Euclidean space
- coordinate systems to solve purely geometric problems by using purely algebraic methods
- geometry as a study not of figures, but of *transformations* (Klein's Erlangen program)[Balbiani et al., 2007] preserving fundamental geometrical properties
- abstract *algebraic* foundation for geometry

From Euclid to Tarski[Aiello et al., 2012] III

Logical foundation of geometry

- *elementary geometry* as first-order theory of Euclidean geometry[Tarski, 1959]
- elementary geometry can be developed axiomatically using just two geometric relations—betweenness and equidistance
- elementary geometry like field of reals via *coordinatisation*
- completeness and decidability of the first-order theory of the field of reals implies an explicit decision procedure for the elementary geometry
- meaning that there exists *algorithm* for deciding the truth of any statement in Euclidean geometry
  - not an efficient one, however

Modal Logics

Non-classical logics

- modelling, analysing, and reasoning about space with non-classical logics
- *modal logics* have proven to be particularly interesting, since
  - they can be more *specific*
  - they have better computational behaviour—very often *decidable*

Modal Logics of Topology I

Distance & metric

- basic problem: *distance*
- approach: notions of *metric*

Topology

- *neighbourhood* as a generalisation of metric
- from neighbourhood to *topology* [Singer and Thorpe, 1967]
- "alongside algebra and geometry, topology became one of the fundamental branches of contemporary mathematics" [Aiello et al., 2012]

Modal Logics of Topology II

Logic and topology

- modal operators reinterpreted
  - $\Box$ *(necessarily)* as the *interior* operator
  - $\Diamond$ *(possibly)* as the *closure* operator of a topological space
- S4[Bull and Segerberg, 1984] is *sound* and *complete* with respect to topological semantics[McKinsey and Tarski, 1944]
- S4 is the *modal logic* of any Euclidean space

Mathematical Morphology [Aiello et al., 2012]

- mathematical morphology (MM) analyses *shape*, *spatial information*, *image processing*
- any mathematical theory dealing with shapes can contribute to MM
- *modal morphologic* from the similarity between the algebraic properties of MM operators and of modal operators
- efficient for spatial reasoning
  - to guide the exploration of space, in a focus of attention process, and for recognition and interpretation tasks

Summary

- we have math and logic tools to *represent*, *analyse*, and *reason* about space
- we can *compute* about space and its organisation with diverse levels of efficiency

Next in Line. . .

Focus on. . .

Distributed Systems: Recap

A new space for software components

- physical distribution of computational systems
  - distribution of *computational units*, *communication channels*, *data*
- local networks building the first virtual spaces
- Internet and IP-based locations for first "global" virtual spaces
- WWW as the first global space shared by agents and humans
  - a *knowledge-intensive* space

Middleware: Recap

Structured environments for software components

- *middleware* as operating systems for distributed systems
  - *software infrastructure* giving structure to *distributed environments*
  - possibly supporting *mobility* [Fuggetta et al., 1998]
- EAI horizontal integration to build *aggregated environments*

Mapping logical distribution upon physical distribution

- middleware
  - provides topological notions for distributed systems
  - maps logical distribution upon physical distribution

JADE provides agent programmers with the topological notions of container and platform to represent *locality* for agents [Bellifemine et al., 2007]

! *sometimes, however, physical distribution is what actually matters*

Distributed Systems & Situated Computations

Situated distributed systems

- physical distribution of computational systems is essential to cope with the distributed nature of many working environments
- . . . as well as with the need for situated computation
  - that is, computations occurring locally where either perception or action are taking place
  - either elaborating on perception, driving action, or both
- essentially, when system requirements mandate for situated computations within a distributed physical environment, situated distributed systems are the only way out
  - e.g., disaster recovery scenarios, environmental monitoring, crowd steering, . . .
- Internet of Things (IoT) just makes the need for situated computation unescapable

Summary

- distributed systems originate from *physical distribution* of computation, communication, and data
- middleware provides logical topology upon physical distribution
- pervasive systems and IoT make situated computation a requirement, and *situated distributed systems* the only viable approach
- nowadays non-trivial computational systems needs to be space-aware

Focus on. . .

Computational Geometry[de Berg et al., 2000]

Computing with geometry to solve problems

- using *space* to *represent problems*—either spatial or non spatial ones
- using *geometry* to make them *computable*
- using *algorithms* to compute solutions

Example

- finding the nearest coffee shop in the campus here in Cesena
- finding it for any point in the campus
- dividing the campus in regions around each coffee shop, including all the points for which each shop is the nearest one
- how do we compute this?

Geographical Information Systems (GIS)

Computing with geography

- representing geographic information
- capturing / storing / checking / displaying data representing positions on the Earth—maybe other planets, too, in the (near) future

→ the notion of space is clearly defined in GIS

- more generally

  A geographic information system is a computer-based system that supports the study of natural and man-made phenomena with an explicit location in space. To this end, the GIS allows data entry, data manipulation, and production of interpretable output that may provide new insights about the phenomena. [Huisman and de By, 2009]

Virtual Reality (VR)

Activity in a virtual environments

- artificial worlds—with artificial *space*
- digitally-created, represented exclusively as *computational environments*
- where "real people" can actually perform sensorimotor and cognitive activity

Definition [Fuchs et al., 2011]

> *Virtual reality is a scientific and technical domain that uses* ***computer science*** *(1) and* ***behavioural interfaces*** *(2) to simulate* ***in a virtual world*** *(3) the behaviour of 3D entities, which* ***interact in real time*** *(4) with each other and with one or more users in* ***pseudo-natural immersion*** *(5) via sensorimotor channels.*

! *here, interaction and immersion are the key concepts*

Gaming

Virtual worlds for gaming

- gaming is the most prominent application of virtual reality
  - e.g., Microsoft Kinect is one of the most well-known applications of VR
- gaming platforms nowadays provide the means for building whole virtual worlds
  - e.g., Unity3D, Unreal Engine

that can be explored from a wide range of diverse devices

Focus on. . .

Location-based Services (LBS)

Adding a virtual layer to physical space

- LBS use information about the position of a user / device to provide information, entertainment, security
  - e.g., Around Me, Mobike, . . .
  - BotFighters, GeoZombie [Prandi et al., 2016], Ingress, Pokémon Go, Harry Potter: Wizards Unite, Minecraft Earth
- *mixed reality* is an old term possibly coming back to use
  - yet, with an uncertain meaning

Augmented Reality (AR)

Integrating virtual and physical space

- merging real-world information with *context-sensitive* digital content in a meaningful way[Furht, 2011]
  - the point here is that the virtual and physical layers mutually affect each other dynamically
- gamification can easily lead to any sort of relevant application—medical, civic, educational, touristic, ...
  - e.g., GeoZombie,[Prandi et al., 2016] a step further (and before) Pokémon Go

Focus on. . .

Altogether Now: Spatial Computing I

When space cannot be abstracted away from computation [Shekhar et al., 2016]

During the "Computing Media and Languages for Space-Oriented Computation" workshop in Dagstuhl (http://www.dagstuhl.de/06361), three classes of spatial systems were identified

- *distributed systems*, where space is either a means or a resource—a.k.a. intensive computing, or *coping with space*
- *situated systems*, where location in space (and time) is essential for computation—a.k.a. *embedded in space*
- *spatial systems*, where space is fundamental to the application problem, is explicitly represented and manipulated, and is essential to express the result of a computation—a.k.a. *representing space*

Altogether Now: Spatial Computing II

Spatial computers, spatial computing [Beal et al., 2011]

- a spatial computer – or, *spatial computing system* – is a computational system where (the logic of) space is essential in representing the problem, define computation, and express the result
- spatial computing is any form of computation where (the logic of) space is relevant to express and perform computation

Spatial Computing Languages (SCL)

Languages for spatial computing

Spatial Computing Languages (SCL)

- were born to address the issue of bringing space *into* programming languages
- allowing programmers to explicitly deal with space-related aspects *at the language level*

Summary

- spatial computing is a sort of "landscape framework" for the many sorts of spatial computations and systems around
- an already-long list of SCL are available, which can be analysed and compared through the conceptual framework [Beal et al., 2013]

Summing Up

- math and logic tools to *represent*, *analyse*, and *reason* about space
- computing about space and its organisation
- different types of code mobility are possible
- spatial computing as a framework for spatial computations and systems

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